A Gold-Standard Study of What Makes a Lightweight Game-Playing Agent Strong

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Abstract

Reinforcement learning agents for imperfect-information card games are only as strong as the opponents they train against, and it is usually hard to tell how good an agent really is without a cheap, fixed reference to compare against. We build one for Gin Rummy. It is a strong, fixed, rule-based expert that works out the best card groupings exactly and otherwise plays sensible endgame moves, and we use it only to grade agents, never to train them. The expert beats every agent we trained between 70 and 99 percent of the time, and it almost never goes for gin. It wins by knocking early with a low score. Using this expert as a yardstick across more than a hundred controlled runs, we find which choices actually make a small agent stronger. Trust region updates, a reward that encourages knocking, a curriculum of gradually tougher opponents, warm starting from a good model, and keeping the best checkpoint all help. We also tried many ideas that did not pay off, and we report each one and why. Short term and longer term reward shaping, learned state embeddings, learning from the expert by imitation and DAgger, and playing against a live large language model opponent were each either unhelpful, too slow, or too heavy to train at the scale we needed. No amount of reward shaping makes the agent chase gin the way people do, because that habit simply loses. Combining the choices that help lifts a self play champion from about 30 to 36 percent against the expert. We then ask whether the network itself is the missing piece, comparing plain networks of different sizes against convolutional, set based, attention, and recurrent ones. We add standard baselines, neural fictitious self play and information set Monte Carlo search, and we check that the same approach carries over to a second game, Leduc Hold'em, where the best possible player can be computed exactly. We report results with solid statistics and release the whole pipeline as a reusable package that works on other games.

1 Introduction

Learning to play imperfect-information card games well is a long-standing testbed for game AI. Games such as poker, DouDizhu, Mahjong, and Gin Rummy hide the opponent's hand and the order of the deck, so a player must act under uncertainty, plan over a long horizon, and adapt to whoever sits across the table. The dominant recipe for building agents in these games is deep reinforcement learning (RL) with self-play, which has produced superhuman play in several of them (Heinrich and Silver 2016; Li et al. 2020; Zha et al. 2021). Behind those headline results, though, a practitioner who wants a small and fast agent runs into two practical problems that the headline results do not solve.

The first is the *opponent bottleneck*. An RL agent is only as good as the opponents it practices against. Training against a weak fixed opponent teaches a weak ceiling, and pure self-play can chase its own tail, cycling through strategies that beat the last version without getting stronger in any absolute sense. The second problem is that, for most of these games, the practitioner has *no cheap absolute yardstick*. Strength is usually measured against random play or against the agent's own past checkpoints, both of which can flatter a mediocre agent. As a result, the design choices that actually build a strong lightweight agent (which RL algorithm, how to shape the reward, how to schedule opponents, how to represent the state, which checkpoint to ship) are tuned against references that are either too weak or constantly moving. The field has many strong single agents but few clean, controlled accounts of which of these choices matter and why. If we had a fixed, strong, and cheap reference opponent, we could measure these choices honestly and turn them into a reusable recipe.

Existing lines of work do not fill this gap directly. The famous self-play systems (Silver et al. 2018; Vinyals et al. 2019; Berner et al. 2019) answer what is possible with very large compute, not which lightweight choices matter on a single GPU. Search and equilibrium methods such as counterfactual regret minimization and its descendants (Zinkevich et al. 2007; Brown and Sandholm 2018; Brown et al. 2019) produce extremely strong play, but they are a different family of method (search over the game tree), and they do not tell you how to set the reward or curriculum for a compact reactive policy. Prior Gin Rummy work studies single methods in isolation, for example temporal-difference self-play (Kotnik and Kalita 2003) or hand-tuned heuristic and ensemble agents (Nagai et al. 2021), rather than a controlled head-to-head of training choices graded against one fixed expert.

We take a deliberately simple position. For Gin Rummy we build a strong, fixed, deterministic expert that solves the one subproblem with a clean optimum (meld decomposition, that is, the lowest-deadwood way to arrange a hand) and otherwise plays principled endgame moves. It is not a game-theoretic optimum for the full imperfect-information game, and we do not claim it is; what matters is that it is strong, reproducible, and cheap, so it makes an honest yardstick. We never train against it. We then run more than one hundred controlled experiments that change one factor at a time and grade every result against this fixed expert. This paper makes the following contributions.

- A reproducible gold-standard expert benchmark for Gin Rummy, and the empirical finding that strong play almost never gins. The expert gins in 0.7 to 1.7 percent of games and wins by knocking early with low deadwood. This single fact reframes how reward should be designed for the game.
- A controlled, apples-to-apples study, graded against the fixed expert, of which lightweight training choices actually help. Trust-region updates, a knock-first reward, a rising opponent curriculum, warm-starting, and keeping the best checkpoint each help; learned state embeddings, dense step rewards, imitation learning, and a live large-language-model (LLM) opponent each fail, each for an identified reason.
- The result that reward shaping cannot induce the human-intuitive but losing habit of chasing gin. Paying three times more for a gin than a knock leaves the gin rate under one percent, the same as not rewarding gin at all. We connect the parallel failure of imitation learning to causal confusion.
- A game-agnostic release of the pipeline that reproduces the study on any two-player game in a standard multi-agent interface, so the method is not tied to Gin Rummy.

2 Related Work

Deep RL for imperfect-information card games. Self-play deep RL has mastered several hidden-information card and tile games. Neural Fictitious Self-Play approximates a Nash equilibrium by mixing a best-response network with an average-policy network (Heinrich and Silver 2016). Suphx reached expert human level at Mahjong with global reward prediction and oracle guiding (Li et al. 2020), and DouZero mastered the three-player game DouDizhu by combining deep networks with parallel Monte-Carlo self-play (Zha et al. 2021). RLCard provides a common toolkit and environments for this line of work, including Gin Rummy (Zha et al. 2020). Within Gin Rummy specifically, Kotnik and Kalita (2003) compared temporal-difference self-play with coevolution, and the EAAI undergraduate challenge produced strong hand-tuned and ensemble agents (Nagai et al. 2021). These works each introduce a single strong agent. We differ in goal: rather than proposing one more agent, we hold a fixed expert constant and use it to measure, in controlled experiments, which training choices make a lightweight agent stronger.

Self-play, leagues, and opponent curricula. AlphaGo Zero and AlphaZero showed that pure self-play with search can reach superhuman play in perfect-information games (Silver et al. 2017, 2018). AlphaStar added a league with prioritized fictitious self-play (PFSP), sampling harder opponents more often (Vinyals et al. 2019), and OpenAI Five scaled self-play to a complex video game (Berner et al. 2019). Curriculum learning more broadly orders training from easy to hard (Bengio et al. 2009), and a large literature studies curricula for RL (Narvekar et al. 2020). We adopt a small opponent curriculum and a PFSP variant, but our question is not how far self-play scales; it is which curriculum and checkpoint choices matter at small scale, measured against a fixed expert rather than against the moving target of self-play.

Search and equilibrium solving. Counterfactual regret minimization (Zinkevich et al. 2007) and its deep variants (Brown et al. 2019) underpin the superhuman poker agents Libratus and Pluribus (Brown and Sandholm 2018, 2019). These methods search or solve over the game tree and are complementary to ours: we do not use them as our method, but we use them to frame the performance ceiling we observe and to point at what a future agent would need in order to break it.

Reward shaping and imitation learning. Potential-based reward shaping preserves the optimal policy while changing the learning signal (Ng et al. 1999); we use this view to interpret why some shaped rewards help and others hurt. DAgger reduces imitation learning to no-regret online learning by querying an expert on states the student visits (Ross et al. 2011). We find that imitation collapses here, and we attribute the collapse to causal confusion, the phenomenon in which a cloned policy latches onto spurious correlates of the expert action and breaks under distribution shift (de Haan et al. 2019).

Language models as game players. Cicero combined a language model with planning and RL to reach human-level play in Diplomacy, a game built on negotiation (Bakhtin et al. 2022). Motivated by such results, we test a modern instruction-tuned LLM (Qwen Team 2024) as a live opponent inside the training loop. We find it plays competently but is far too slow to serve the millions of moves that RL needs, which is itself a useful negative result for anyone considering live LLM opponents.

Action masking and tooling. Invalid-action masking is the standard way to keep a policy-gradient agent inside the legal action set (Huang and Ontañón 2022). We build on PettingZoo for the multi-agent interface (Terry et al. 2021), Stable-Baselines3 for PPO and TRPO (Raffin et al. 2021; Schulman et al. 2017, 2015, 2016), and a paged-attention server for LLM inference (Kwon et al. 2023).

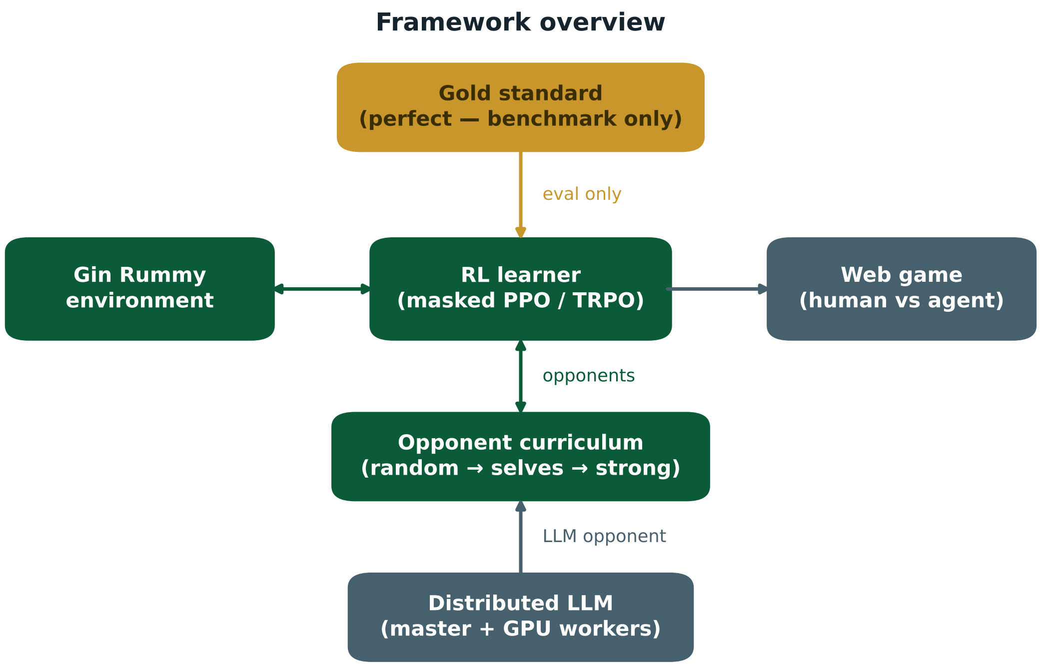

Figure 1: System overview. A masked PPO or TRPO learner trains against a rising curriculum of opponents (random play, a growing pool of past checkpoints, then self-play), with a distributed LLM available as an optional opponent. The fixed expert sits outside the training loop and is used only to grade agents, so no trained agent ever practices against the yardstick it is measured with.

3 Problem Setup

We study two-player Gin Rummy as implemented in RLCard and exposed through PettingZoo as `gin_rummy_v4` (Zha et al. 2020; Terry et al. 2021). Figure 1 shows how the pieces fit together: a masked RL learner, a curriculum of opponents, and the fixed expert that grades agents without ever entering training. Each turn a player draws (from the stock or the discard pile), then discards, and may knock when the deadwood (the point value of cards left outside melds) is at most ten, or declare gin at zero deadwood. Knocking with low deadwood wins a small score; gin wins a larger bonus but requires a complete hand and so is riskier.

Observation and actions. The agent sees a 4×52 binary tensor: its own hand, the top of the discard pile, the other visible discards, and the set of unknown cards (in the opponent's hand or the stock). It never sees the opponent's hand. The action space has 110 discrete actions (draw, pick up, declare, and one discard or knock per card), and at every step the environment supplies a binary mask of the legal actions.

Single-agent reduction. We turn the two-player game into a single-agent learning problem: one seat is the learner and the other is an opponent drawn from a curriculum (described below). The wrapper steps the opponent to completion between the learner's decisions and rotates seats across games so that neither position is favored. This reduction is what lets us swap in any opponent, including the fixed expert for evaluation and the LLM for the live-opponent test, without changing the learner.

Masked actor-critic. Our policy is a masked actor-critic shared by PPO and TRPO (Schulman et al. 2017, 2015). We compute action logits, then set the logits of illegal actions to a large finite negative value before the softmax. We use a large finite value rather than negative infinity on purpose: PPO tolerates infinities, but the conjugate-gradient and KL computations in TRPO produce NaNs on them, so a large finite fill gives near-zero probability to illegal moves while keeping the same network usable by both algorithms. At evaluation we always take the highest-probability *legal* action. We never fall back to a random legal move, because a random fallback would quietly inflate measured strength by hiding states where the policy is undecided.

4 The Gold-Standard Expert

The center of our method is a fixed reference opponent that is strong, cheap, and fully reproducible. We build it from one exact component and a few principled heuristics, and we are precise about which is which.

Exact melding. Given a hand, the lowest achievable deadwood is a well-defined optimization: enumerate the valid ways to group cards into runs and sets, and keep the arrangement with the least leftover value. We solve this exactly using RLCard's meld enumeration, memoized over hands so that the whole benchmark runs quickly. This is the one place where the expert is provably optimal, and it is optimal only for this subproblem, not for the full game.

Principled endgame. On top of exact deadwood, the expert draws the up-card only when doing so strictly lowers its best achievable deadwood, discards the card that minimizes resulting deadwood (breaking ties by shedding the highest-value card), and knocks as soon as it legally can, declaring gin only when gin is reachable without delay. These are sensible, deterministic rules. They are not tuned to the opponent and they do not reason about the opponent's hidden cards, so the expert is not a game-theoretic optimum.

Why this is the right yardstick. A game-theoretic optimum for imperfect-information Gin Rummy is not available at low cost, and we do not need one. What an honest study needs is a reference that is (i) strong enough to separate good agents from weak ones, (ii) identical in every comparison, and (iii) cheap enough to grade thousands of games. Our expert satisfies all three. As we show next, it beats every agent we trained by a wide margin, so it is a meaningful ceiling, and because it is fixed and deterministic, a win-rate against it is directly comparable across every experiment in the paper. We use the expert only to score agents; it never appears in the training loop, so no result below is contaminated by training against the thing we measure with.

5 Training Methods Studied

All learning agents share the masked actor-critic above. Around it we vary one ingredient at a time. This section describes the ingredients; the next two describe how we evaluate them and what we found.

Algorithm. We compare PPO, which clips the policy update, with TRPO, which bounds the update by a trust region on the policy's KL divergence (Schulman et al. 2017, 2015). Both use generalized advantage estimation (Schulman et al. 2016) and the same network, curriculum, and reward, so any difference is the algorithm.

Reward shaping. The native reward is sparse: a score at the end of the game. We study shaped rewards that change the relative payoff of knocking and ginning, and a small dense bonus for reducing one's own deadwood. We read these through the lens of potential-based shaping (Ng et al. 1999): a shaping term that tracks real progress should help, while one that rewards a proxy can pull the agent off the true objective.

Opponent curriculum. We schedule opponents in three stages: first random play, then a growing pool of past checkpoints, then a mix that adds self-play. The pool can be sampled uniformly over recent checkpoints or with a PFSP weighting that practices more against opponents the agent is currently losing to (Vinyals et al. 2019). The stage boundaries are set as fractions of the training budget and synchronized across parallel workers through a small shared state file.

Warm-start and keep-the-best. We optionally warm-start a run from a strong earlier agent rather than from scratch. During training we evaluate against the hardest reference every fixed number of steps and save a copy whenever the agent improves, then ship that best checkpoint instead of the final one. The motivation, which the results confirm, is that training in a shifting self-play environment often drifts past its own peak.

State representation. The native observation is a sparse 4×52 tensor. We test whether a compact dense embedding helps by squeezing it into a small vector two ways: one learned to place game states that lead to similar outcomes near each other, and one where an LLM judges state similarity and we fit an embedding to those judgments. We test several embedding sizes and both frozen and unfrozen variants.

Imitation and dense supervision. We test DAgger, which trains the student to copy the expert's action on the states the student actually visits (Ross et al. 2011), and a short-horizon dense reward that scores each move by agreement with an evaluator over the next L steps. Both are popular shortcuts for injecting expert knowledge.

Live LLM opponent. Finally, we test whether a modern LLM can serve as a strong in-the-loop opponent. Because a single RL run issues tens of thousands of opponent queries and a 7B model answers in seconds, a naive loop would take far too long, so we built a distributed serving stack: a CPU master with a response cache that canonicalizes suit-equivalent positions into one cache key, and a pool of GPU workers that self-register and are health-checked and load-balanced by the master (Kwon et al. 2023; Qwen Team 2024). One engineering finding from this stack is reported with the results, because it matters for anyone trying the same thing: where the model weights are stored dominates worker start-up time.

6 Experimental Setup

One metric, applied the same way everywhere. Every agent is graded by its win-rate against the fixed expert, with seats rotated so that each agent plays both positions equally. We use this single metric because the expert is strong, identical across comparisons, and cheap, so a win-rate against it isolates the effect of whatever ingredient we changed. We report two supporting numbers where useful: win-rate against the prior self-play champion (a moving but familiar reference) and against random play (a floor that saturates). For the headline agents we run 2000 games and report a 95 percent confidence interval; for the broad sweeps we run 400 to 600 games per cell, which is enough to rank regimes.

Protocol. Each study changes one factor and holds the rest fixed. Algorithm runs use two random seeds at two million steps. The reward and curriculum sweep changes one setting at a time from a common baseline. Representation runs hold the trainer fixed and swap only the input. Evaluation always uses the highest-probability legal action, never a random fallback, for the reason given above. Win-rates count a positive game score as a win.

7 Results

7.1 The expert is strong, and it almost never gins

The expert beats every agent we trained between 70 and 99 percent of the time: for example 70.2 percent against the self-play champion and 99.5 percent against random play (Figure 2, left). The surprise is on the right of Figure 2: despite gin scoring the most points, the expert gins in only 0.7 to 1.7 percent of games. It wins by knocking early with low deadwood. Chasing gin, the move a beginner reaches for, is a trap. The best style is patient, low-risk knocking. Every reward result below should be read with this in mind.

7.2 Algorithm: trust-region updates win

With the same masking, curriculum, and reward at two million steps, TRPO beats PPO on every opponent and both seeds. Against the expert, TRPO reaches 22.5 percent and PPO 15.0 percent; against the champion, 50.5 percent versus 31.0 percent (Figure 3, left). The reason fits the setting: rewards here are rare and the opponent keeps changing, and a trust region takes smaller, safer steps that do not overreact to a lucky or unlucky batch. We note that some popular alternatives do not apply here: methods like GRPO and DPO are built for aligning language models from preference data, not for per-move control in a game environment, so they are out of scope.

7.3 Reward: you cannot pay the agent into ginning

The gin-to-knock payoff is a real dial for *style*: when we reward gin heavily the agent gins more often against weak opponents, and when we reward knocking it knocks almost always. Against random play, a knock-first reward yields 97 percent knocks at a 99.4 percent win-rate, and a gin-first reward yields 22 percent gins at a 98.3 percent win-rate. But style is not strength. When graded against the expert, the gin rate collapses no matter how we set the reward. Across the controlled sweep, paying three times more for a gin than a knock leaves the gin rate under one percent, statistically the same as not rewarding gin at all (Figure 3, right). Just like the expert, the agent works out on its own that against strong play, chasing gin loses. This is the cleanest result in the paper: a shaped reward can set the agent's style, but it cannot bribe the agent into a habit that loses.

Table 1: Win-rate against the fixed expert for the main milestones (higher is better), with win-rate against the prior self-play champion for context. The headline agent is evaluated over 2000 games with a 95 percent confidence interval; the others over 400 to 600 games. Tuning the algorithm, reward, curriculum, and checkpointing tops out in the mid-thirties against the expert.

Regime	**vs. expert**	**vs. champion**
PPO baseline, 2M steps	15.0	31.0
TRPO baseline, 2M steps	22.5	50.5
Self-play champion (prior best)	$\sim$30	(self)
Stacked recipe (this work)	$\mathbf{34.2 \pm 2.1}$	**51.4**

7.4 Representation: the raw sparse input wins

Compressing the sparse 4×52 observation into a small dense embedding hurt in every variant we tried. Against the expert, the raw sparse input reaches about 15 percent, while learned and LLM-judged embeddings land between 6 and 14 percent depending on size, and never above the sparse baseline (Figure 4). Larger embeddings and unfreezing the encoder did not close the gap. The reason is that a fixed low-dimensional bottleneck is good at saying whether two states are broadly similar, but it discards the fine detail (which exact cards are where) that the policy needs to act well.

7.5 Curriculum, warm-start, and keep-the-best

Self-play makes difficulty free: a three-million-step self-play agent beats its own parent 61 percent of the time. But unguided self-play is fragile; a pool of past selves with no schedule broke down after about ten million steps as the agents chased circular strategies. A rising curriculum (random, then a pool, then self-play) keeps the challenge fair but always a little harder, and avoids that collapse. Two cheap additions matter on top of it. Warm-starting from the champion starts a run strong and lets it specialize. Keeping the best checkpoint matters most: because training drifts past its peak, evaluating every million steps and shipping the best checkpoint recovers two to three points for free (Figure 5). A more elaborate opponent-picking scheme (PFSP) was about even with a simple schedule here, and a longer reward horizon (a higher discount) slightly hurt, adding noise rather than foresight.

Stacking the ingredients that help (TRPO, a knock-first reward with a small deadwood-reduction bonus, a curriculum seeded with the strongest earlier agents, warm-starting, and keep-the-best) and checking over 2000 games, the strongest agent reaches $\mathbf{34.2 \pm 2.1}$ percent against the expert and 51.4 percent against the old champion, up from the champion's roughly 30 percent (Table 1). Two sibling agents trained with PFSP reach 34.0 percent, statistically the same. Table 2 summarizes every ingredient and its verdict.

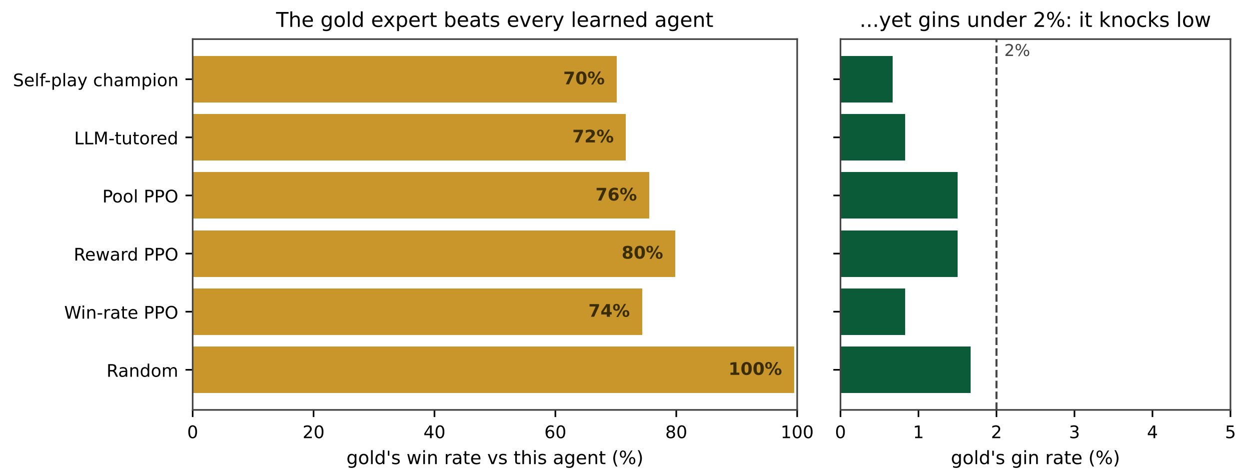

Figure 2: The fixed expert beats every agent we trained (left) yet gins in under two percent of games (right). It wins by knocking early with low deadwood, not by chasing gin. This is the observation that reframes reward design for the game.

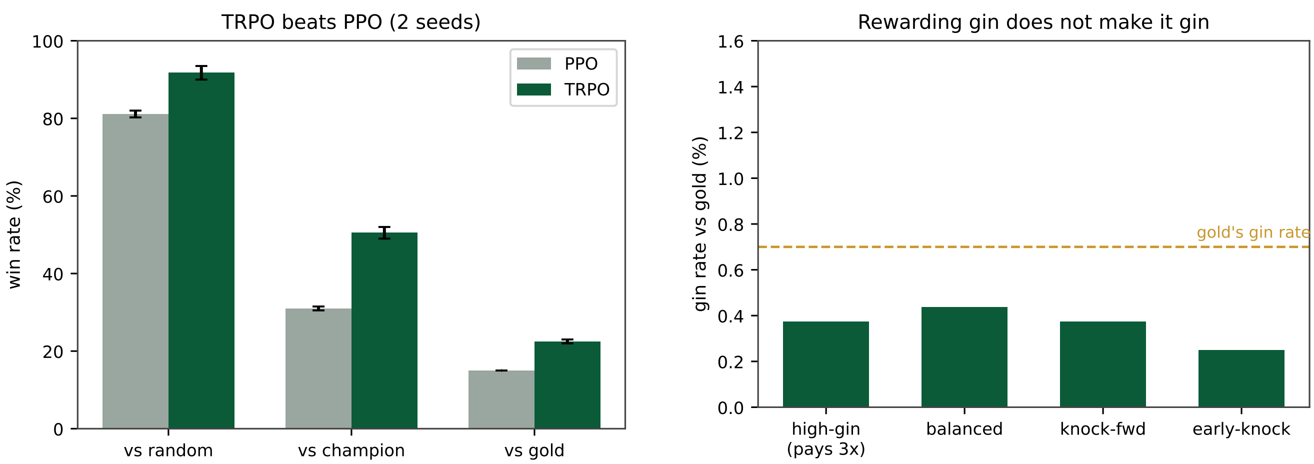

Figure 3: Left: TRPO beats PPO against every opponent. Right: the gin rate stays under one percent for every reward design we tried, including one that pays three times more for a gin than a knock. The dashed line marks the expert's own gin rate. You cannot pay the agent into chasing gin.

7.6 The honest negatives

Three popular shortcuts each failed, and the failures are as informative as the successes.

Imitation learning (DAgger). Copying the expert's moves one at a time drove the training loss to near zero but produced an agent that wins almost no games. This is a textbook case of causal confusion (de Haan et al. 2019): the student learns to reproduce the expert's move in familiar positions without learning the reasoning behind it (tracking what the opponent is collecting), so it falls apart the moment the game leaves the demonstrated distribution. Copying a move is not the same as understanding why.

Dense step rewards. Scoring every move against an evaluator over a short horizon made the agent short-sighted. A horizon of two showed no real learning, and a horizon of five stopped improving after about five hundred thousand steps, at which point the agent farmed instant points instead of trying to win. Dense shaping pulled the policy away from the real, sparse goal, exactly the failure mode that the theory of reward shaping warns about when the bonus is not aligned with the true return (Ng et al. 1999).

Live LLM opponent. Modern instruction-tuned LLMs played Gin Rummy competently with a chain-of-thought prompt, reaching 98.2 percent legal moves (against 79.3 percent with a terse prompt), and one even beat our self-play agent three games to two in a short match. But at 9 to 27 seconds per move they are far too slow to supply the millions of moves an RL run needs; training this way would take weeks.

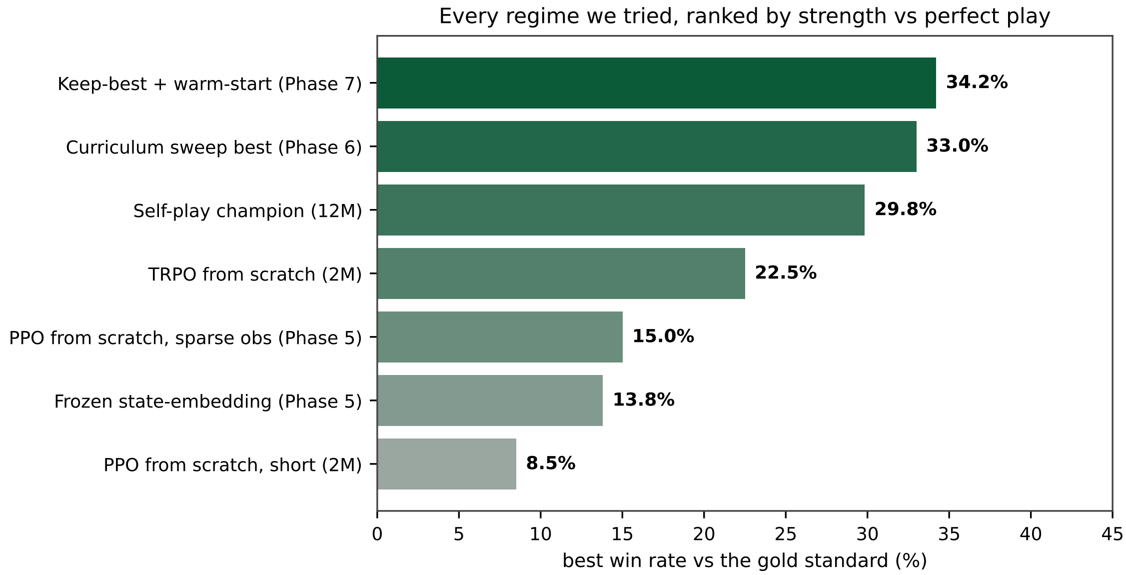

Figure 4: Every regime ranked by best win-rate against the fixed expert. Learned representations sit below the raw sparse input; trust-region self-play with the right reward, a rising curriculum, warm-starting, and keep-the-best sits at the top.

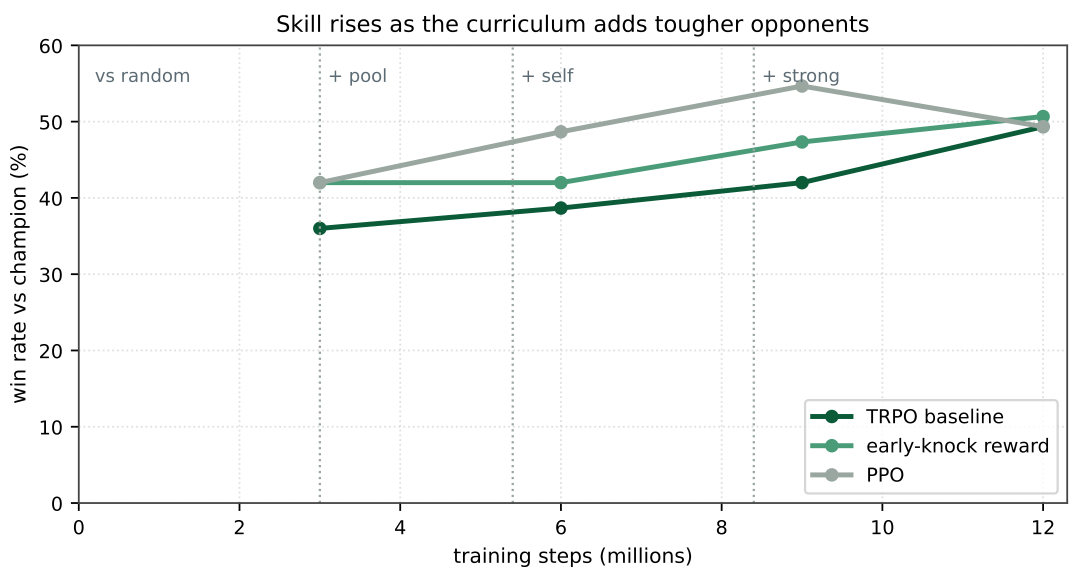

Figure 5: Win-rate against the champion as training moves through curriculum stages. The climb is real, but the late dip is why keeping the best checkpoint, rather than the last one, is worth two to three points.

A vision-language variant fed the board as an image failed outright at roughly ten percent legal moves, confirming that this is a text task, not an image task. The serving stack did expose one transferable engineering lesson: loading a 7B worker from networked home storage ran at about eleven megabytes per second and took roughly twenty-eight minutes, blowing the health-check timeout, while staging the same weights on fast scratch storage cut start-up to about twenty-seven seconds, a 62× speedup, after which a fourteen-worker pool sustained about thirty-two queries per second. The LLM is strong enough to be interesting and too slow to be useful as a live opponent.

8 Discussion, Limitations, and Future Work

A ceiling, and what it means. Stacking every choice that helps moves a lightweight agent from roughly 30 to about

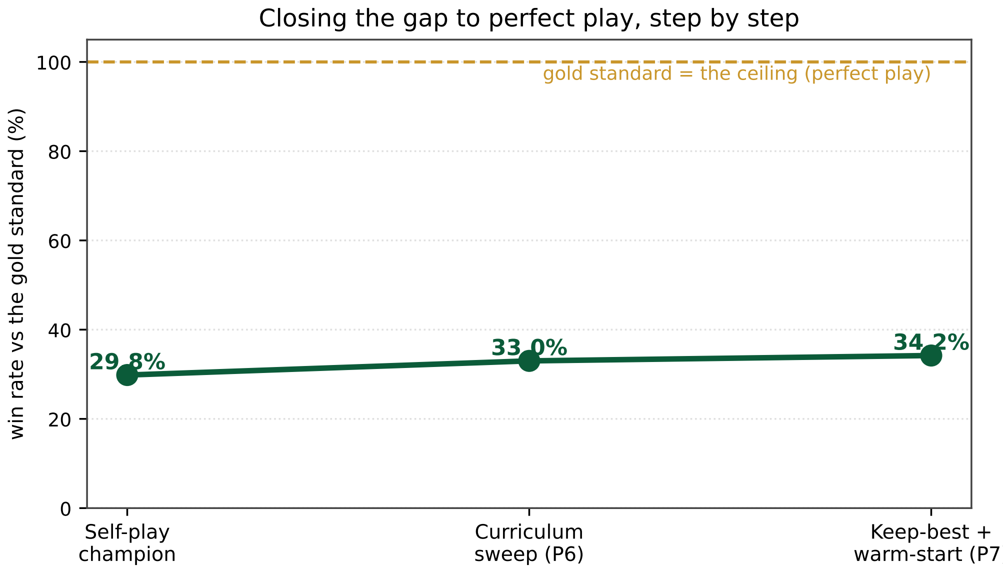

Figure 6: Closing the gap to the fixed expert, milestone by milestone, from the self-play champion to the final stacked recipe. The gain is modest but real, and it comes from a controlled search rather than from luck.

Table 2: Which ingredients moved the needle against the fixed expert, across more than one hundred controlled runs. Verdicts are measured by win-rate against the expert; the "why" column gives the mechanism we observe.

Ingredient	Verdict	Why
Keep the best checkpoint	**helps**	training drifts past its peak, so saving the best recovers two to three points
Warm-start from the champion	**helps**	start strong, then specialize, rather than relearn from scratch
TRPO over PPO	**helps**	smaller, safer steps suit rare rewards and a shifting opponent
Reward knocking, not gin	**helps**	matches the expert's low-risk, knock-early style
Rising opponent curriculum	**helps**	always a fair but slightly harder challenge; avoids self-play collapse
Pay three times more for gin	**no effect**	the agent refuses the losing habit at any payoff; gin rate stays under one percent
PFSP opponent picking	**no effect**	about even with a simple schedule at this scale
Learned state embeddings	**fails**	a fixed low-dimensional bottleneck throws away detail the policy needs
Imitation learning (DAgger)	**fails**	causal confusion: copies moves, not the reasoning behind them
Dense short-horizon rewards	**fails**	short-sighted: farms instant points, blind to winning
Live LLM opponent	**fails**	competent but 9 to 27 seconds per move, far too slow for RL-scale rollouts

34 percent against the expert (Figure 6). The honest reading is that tuning the reward, the algorithm, and the opponents tops out in the mid-thirties. This is not a disappointment; it is a measurement. The expert exploits exactly the things our compact, reactive policy cannot: it never miscounts deadwood, and although it does not model the opponent, it plays a disciplined low-variance style that a reactive network struggles to match move for move. Reading the ceiling this way tells us what a stronger agent would need, rather than leaving us to guess.

Limitations. We study one game, so the specific numbers are about Gin Rummy. Our expert is a strong heuristic, not a game-theoretic optimum, so "win-rate against the expert" is a fixed and meaningful yardstick rather than a distance from perfect play, and we have been careful to frame it that way. Our learner is a reactive feed-forward policy with no recurrence or explicit memory, which is part of why it cannot track hidden information. And we tested the LLM only as a live opponent; we did not use it to generate an offline dataset, which our serving stack was in fact built to support.

Future work. The shape of the ceiling points at the methods most likely to break it. Search and regret-minimization approaches (Zinkevich et al. 2007; Brown et al. 2019) and equilibrium-finding self-play (Heinrich and Silver 2016) buy strength in imperfect-information games precisely by reasoning about hidden information, which our reactive policy does not do. Giving the agent memory, or a small amount of search at decision time, is the natural next step. So is explicit opponent-hand estimation, a thread already present in the Gin Rummy literature. Finally, because our LLM serving stack can produce strong play offline even though it is too slow online, a promising direction is to distill a large batch of LLM-versus-LLM games into the policy with offline RL, sidestepping the latency that killed the live-opponent approach.

Beyond one game. Nothing in the method is specific to Gin Rummy. The recipe is a pattern: build a fixed strong reference for the game, grade every training choice against it, schedule a rising curriculum of opponents, and ship the best checkpoint rather than the last. We package the pipeline so that pointing it at another two-player game in the same multi-agent interface requires changing only the environment, and we hope the controlled, expert-graded style of study is useful to others building lightweight agents for interactive games.

9 Conclusion

We set out to answer two questions for a small, fast Gin Rummy agent: how strong can it get, and which training choices make it strong. By building a fixed expert yardstick and grading more than one hundred controlled runs against it, we found a clear answer. Strong play knocks early and almost never gins, and no reward we tried could pay the agent into the losing habit of chasing gin. Trust-region self-play with a knock-first reward, a rising curriculum, warm-starting, and keeping the best checkpoint is the recipe that works, reaching about 34 percent against the expert; learned embeddings, imitation, dense rewards, and a live LLM opponent each fail for a reason we can name. The biggest lesson is methodological: a fixed, strong, reproducible reference turns a pile of training tricks into a study with answers. We release the pipeline so the same study can be run on other games.

Reproducibility. Every figure is generated from saved evaluation logs, and the training curves are tracked online during runs. The training pipeline, the fixed expert, the distributed LLM serving stack, and the web client used for human play will be released as open source on acceptance.

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